



Fall 2026

CISC/CMPE 320 : Fundamentals of Software Development

1

Topic: Introduction

Welcome!

- Fundamentals of Software Development Fall 2026
 - Mondays 4:30 – 5:30
 - Wednesdays 3:30 – 4:30
 - Fridays 2:30 – 3:30
- Course website: OnQ

Instructor

- Dr. Samir Mohammad
- Office: Sutherland 532
- Email: [sm373 \[at\] queensu.ca](mailto:sm373@queensu.ca)
- Office hours: Tuesdays 2 – 3
Wednesdays 12 – 1

Preferred to have an appointment

Contact your instructor Instruction

To communicate with the instructors. Please use the following format for your “subject” of the email:

320-1234 -Topic

The subject contains 3 parts separated by underscore “_” :

- 1) the “320” part stands for the course number,
- 2) the “1234” part represents last four digits of your student ID,
- 3) the third and last part contains the topic of your email.

Please insert “_” Underscore between the three parts

Example: A student ID # is 20143670 and s/he are sending an email regarding “Assignment 2” then the subject of the email should be:

320-3670-Assignment 2

TAs

Ref.	Name
G1	Mohammad Faiyaz Khan
G2	Adekunle Akinjobi Ajibode
G3	Mohammed Makarem
G4	Jihyeon Park
G5	Ali Sedigh Moghadam

- TA Office Hours With the Tutorial :
 - Mondays 9:30 – 11:30
 - Thursdays 9:30 – 11:30
 - Thursdays 12:30 – 2:30

Overview

- C++ course
- Discussion of several software engineering topics will aid in the completion of a team project.

Grading scheme

- **Grading scheme**
 - 30% four assignments, (due roughly every three weeks starting week 3)
 - 35% one project
 - 35% one final exam
- **Assignments**
 - Electronic submission through onQ.
 - One assignment submission per person.
- Use `ISO C++` and you can use any `IDE` for the `GUI`.

Assignment rules and deadlines

- Assignments marks and comments will be uploaded to OnQ by your TA.
- Each assignment will have
 - One week for **deadline** and an
 - **Emergency Period** of one week.
 - 7 days Extension (Emergency Period) is given to all students including extensions for accommodations and short-term considerations.
 - See Syllabus for details.

Assignment rules and deadlines

“Group Efforts”:

- I encourage you to discuss your difficulties with your peers and your TA.
- You may review and learn from other people’s code to explore alternatives or to seek help in a certain area.
- **But, you cannot copy other people’s code!**

Location and Timing of Final Examinations

- **Students should delay finalizing any travel plans until after the examination schedule has been posted.**
- **Exams will not be moved or deferred to accommodate students , travel/holiday plans or flight reservations.**
- As indicated in Academic Regulation 8.3, students must write all final examinations in all on-campus courses on the Kingston campus.

C++ Team Project

- Read more details on OnQ.
- You will be assigned to agile teams by Head TA.
- Each team member should write code and participate fairly.
- Students will be divided into groups during the 1st Tutorial timem.
- **Right now:**
 - You must attend the first Tutorial in order to be joined a group and proceed with your role.
 - read more about the course project on OnQ
 - Start thinking about possible projects.
- You should know what team you are in during the first Tutorial.

Team Meetings

- You can meet in-person and using Microsoft Teams.
- A meeting needs to happen **at least once a week**
(Weekly Tutorial)
- Tutorial time of the course is dedicated for groups meet. However, other meeting-time can be scheduled according to the convenience of the team's members, if needed.
- Your mentor will meet with you in regular basis
(Weekly during the tutorial)

C++ Team Project, Cont.

- I will expect some leadership from those of you with C++ experience and teamwork experience.
- However, other team members are expected to help with the project
 - You are expected to start writing advanced code with tools that are provided for you in the CASE tool.
 - Members of a group can support each other and share knowledge and experience. So, all members would be familiar with other members contributions.
 - It is normal to exchange comments with your peers, whether it is positive or negative. Engage yourself in a discussion to defend your idea and/or accept your colleagues' comments, so that your discussions will lead to a consensus. **Exactly like what would happen in a real job**

Grading the project

- Your team will present your project for grading, in week 12 .
- You will create two reports:
 - 1) Requirement Analysis Document (or “**RAD**”) in week 7 and
 - 2) System Design Document (or “**SDD**”) in week 9.
- You must use the supplied CASE tools.
- Your team will maintain careful record on the CASE tools.
Your part would be to maintain your own page -a “diary”- listing what you have completed and when.
- Your team will be assigned an initial grade.
- Your individual grade will be affected negatively if you don’t contribute fairly to the project.

Grading the project, Cont.

- See the OnQ page for a more detailed breakdown of how marks are assigned to the project. See rubrics of RAD and SDD.
- Attendance of tutorial and your contributions on each meeting is graded by the TA. This grade is part of your project's overall grade.

CASE Tool

- CASE tools. (What does “CASE*” stand for?)
Jira, Confluence, Bitbucket, Slack, MS Teams.
(This year we will search for a different CASE tool,
see next slides)

* CASE: Computer-Aided Software Engineering

Jira- Previous Years

- Used to have our own datacenter space for this course.
 - This system gave each team
 - a SCRUM* planning tool (**Jira**),
 - a document creation and sharing tool (**Confluence**), and
 - a git repository** (**Bitbucket**), and
 - a chat tool (MS teams or Slack)-each of which you must use.
 - Other chat services to discuss the project. (like FB messenger) is not recommended
-

* Scrum is an agile project management framework

** This repository tracks all changes made to files in your project, building a history over time.

Jira-Previous Years , Cont.

- Jira use is a must:

Discussions should only happen in the CASE tools provided

Course Goals

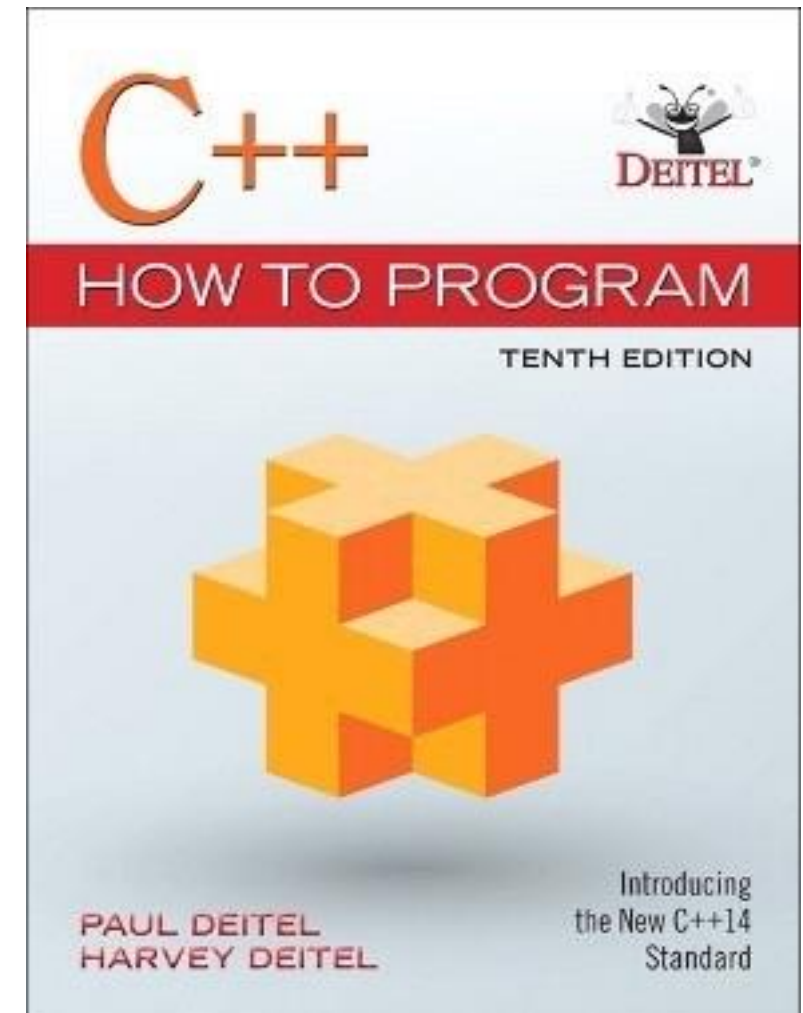
- To give you solid proficiency in ISO standard C++. Expose you to coding in C++/CLI.
- To give you some experience about what it is like to develop software as part of a team.
 - ─ Learn some “software carpentry” and software engineering.
- *All good stuff for your resume!*

Learning C++

- If you have not coded in C++, your Java experience will help.
- C++ is not a “safe” language like Java - possibilities to face problems are higher than that of other languages.
- Even if you are strong in C++, it does not hurt to reinforce the fundamentals.

Resources

- Recommended textbook:
- More good web links, too-
see the Resources section on OnQ.
 - <https://www.learncpp.com/>
 - <http://www.cppreference.com/>



Weekly Schedule

Week	Of	Subject	Assignments Due 23:59 on Wednesdays	Projects Due Fridays
1	Sep 1	Introduction, Setup, History, Basics		
2	Sep 8	Data Types, Pointers, Arrays		
3	Sep 15	Agile, Passing, Operators, Loops, String	Assign 1 due (Sep 17, 11:59 pm)	
4	Sep 22	I/O, Classkey, Exceptions, RAD_SCRUM		Project Outlines (Sep 26, 11:59 pm)
*5	Sep 29	Functions, Overloading 1	Assign 2 due (Oct 1, 11:59 pm)	
6	Oct 6	Overloading 2, Overloading 3, Repositories, SDD		
Oct 14 - 18 R e a d i n g W e e k				
7	Oct 20	SW Analysis 1, SW Analysis 2, SW Design.		RAD Submission (Oct 24, 11:59 pm)
8	Oct 27	Namespaces, Memory, Memory Errors, Help Tools		
9	Nov 3	Inheritance 1	Assign 3 due (Nov 5, 11:59 pm)	SDD Submission (Nov 7, 11:59 pm)
10	Nov 10	Inheritance 2, Inheritance 3, Inheritance 4, Testing.		
11	Nov 17	Testing 1, Testing 2, Templates, STL	Assign 4 due (Nov 19, 11:59 pm)	
12	Nov 24	Smart Pointers		Project Presentation Nov 28
13	Dec 1	Review Dec 2 is end of Fall term		

CISC/CMPE 320 - Fall Schedule 2026																					
	8	8:30	9	9:30	10	10:30	11	11:30	12	12:30	1	1:30	2	2:30	3	3:30	4	4:30	5	5:30	
Mon				(1) 30 + 18 = 48 (6 groups)																	
Tue																					
Wed																					
Thur				(2) 26 + 22 = 48 (5 Groups)						(3) 41 + 4 = 45 (4 Groups)											
Fri																					

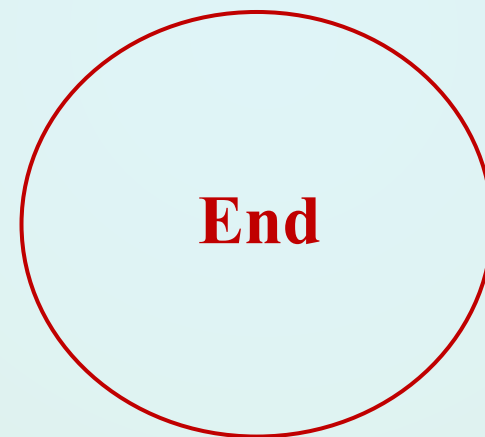
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Students to be moved

From: Thursday 12:30 - 2:30

To: Monday 9:30 - 11:30

No.	Student Name	Student ID
1		
2		
3		
4		
5		
6		
7		



Next: Getting Setup

